

Engineer Cradle: LLM-Driven Digital Twin System for Personalized Learning and Lab Management in EEE

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Abstract

Emerging Engineering Education (EEE) is crucial for training engineering and technical personnel, and colleges are actively building student practice bases. However, new engineering education requires interconnection across multiple disciplines and has issues

such as the contradiction between strict laboratory equipment management and the independent learning needs of students. Utilizing the rich data from our university's Emerging Engineering Research & Development Building, we have constructed an intelligent agent using large models and a digital twin system of the building to provide personalized guidance for student academic planning and equipment management. We have also developed a multi-agent framework to generate learning path plans and course equipment suggestions, supporting multi-round dialogue iterative learning. We developed the digital twin system with unity and the digital human system with Motionverse. The demonstration presents a solution for new engineering education and its application potential.

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CCS Concepts

• **Applied computing** → *Interactive learning environments*; • **Human-centered computing** → *Ubiquitous and mobile computing*.

Keywords

Digital Twin, Digital Human, Large Language Model, Emerging Engineering Education

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1 Introduction

Emerging Engineering Education (EEE) has played an important role in training a large number of engineering and technical personnel for all engineering faculty and students, and local universities have successively begun to build relevant student practice bases to ensure the smooth conduct of emerging engineering teaching, practical and innovative activities[13]. During the implementation of these activities, it was found that new engineering courses have the characteristic of requiring interconnectivity across multiple disciplines and fields[15]. This demands that in the education and training process, content from multiple disciplines and fields needs to be serialized and organized in terms of information. Meanwhile, to cultivate relevant technical personnel for the new engineering education system, universities have spared no effort in equipment acquisition[8]. The increasing variety and quantity of equipment have also laid a solid foundation for students' personalized and diverse learning and research needs.

However, engineering education in current universities is rather structured[1]. Such a learning mode is effective in shaping standardized and qualified engineers, but allows little room for students under the same program to achieve diverse individual goals. To enable students to engage in self-directed learning and experiments based on their career plans and learning needs, educators need to identify suitable learning paths to prevent students from taking detours. Meanwhile, experimental equipment is subject to strict regulations and approval procedures to ensure users' safety. The conflict between the strict equipment management in practice bases and students' self-learning and innovation needs remains a persistent issue[6, 16]. For new learners, they are often at a loss about the available equipment for their project tasks in the absence of teachers' guidance, leading to the idling of valuable experimental resources and students missing out on practical opportunities.

Fortunately, the emergence of large language models (LLMs) has offered new approaches for retrieving and organizing massive amounts of information and knowledge, which can integrate knowledge across multiple fields of emerging engineering education for comprehensive analysis[2]. Also, persona-based Q&A applications can present information from LLMs in a more engaging way, improving learners' comprehension[5]. Meanwhile, the maturation of digital twin technology, with practical experience in architecture and factories[11, 12, 14], has provided more efficient methods for laboratory equipment management and safety control.

Therefore, leveraging our university's Emerging Engineering Research & Development Building's vast information resources,

we've developed a large model based intelligent digital human and a digital twin system. This offers crucial support for students' new engineering academic planning and manager's laboratory equipment management. Through the digital twin system, teachers and students can understand the building in details. Manager can remotely monitor and control its operation with the intelligent agent's assistance. Most importantly, students can have the intelligent agent create personalized learning plans based on their learning goals and academic performance. These plans are linked to the digital twin system, enabling students to locate recommended courses and equipment within it, thus facilitating self-directed learning and research.

However, current single-agent approaches, limited by context window size, struggle to handle extensive student information and diverse managerial queries. To address this, we've developed a multi-agent framework. It uses a triage-based classification and follows a "profile"- "plan generation"- "reflection"- "iteration" route to create personalized student learning paths and equipment usage recommendations. Additionally, it supports multi-round dialogue iterations to refine and generate more student-centered learning plans.

Finally, we developed the digital twin system with Unity[10] and the digital human system for the intelligent agent with Motionverse[4]. The demo presents our solution for the multi-interdisciplinary, theory-practice integration features of Emerging Engineering Education, showcasing its potential in school education, production training, etc.

2 Digital Twin SYSTEM

As shown in Fig.1, our system consists of a digital twin system and an intelligent agent digital human system. The digital twin system generates multimodal information related to the practice base, which serves as reference data for the intelligent agent digital human's information-based decision making. This allows the digital human to generate appropriate responses to different user queries.

2.1 Building of Digital Twin System

We have developed a digital twin system to efficiently and clearly present the relevant information of the Emerging Engineering practice base. During building modeling, we applied high-precision 3D scene reconstruction to the exterior using oblique photography, feature extraction, matching, aerial triangulation, dense point cloud generation, and texture mapping. For interior scenes, we used CAD drawings for detailed modeling. Then, we performed sensor data transmission. For cameras, we obtained video streams via the RTSP protocol[7], decoded and rendered them with a Unity3D plugin for real-time monitoring. For other sensors, we sent data to the Unity3D platform via the MQTT protocol[3] for real-time environmental data monitoring.

The system scene, rendered in Unity 3D, is both efficient and visually appealing. It enables real-time monitoring and remote control of the practice base's status, including infrastructure and lab equipment operation. This allows managers to fully understand the engineering building's status without being on site. Also, the

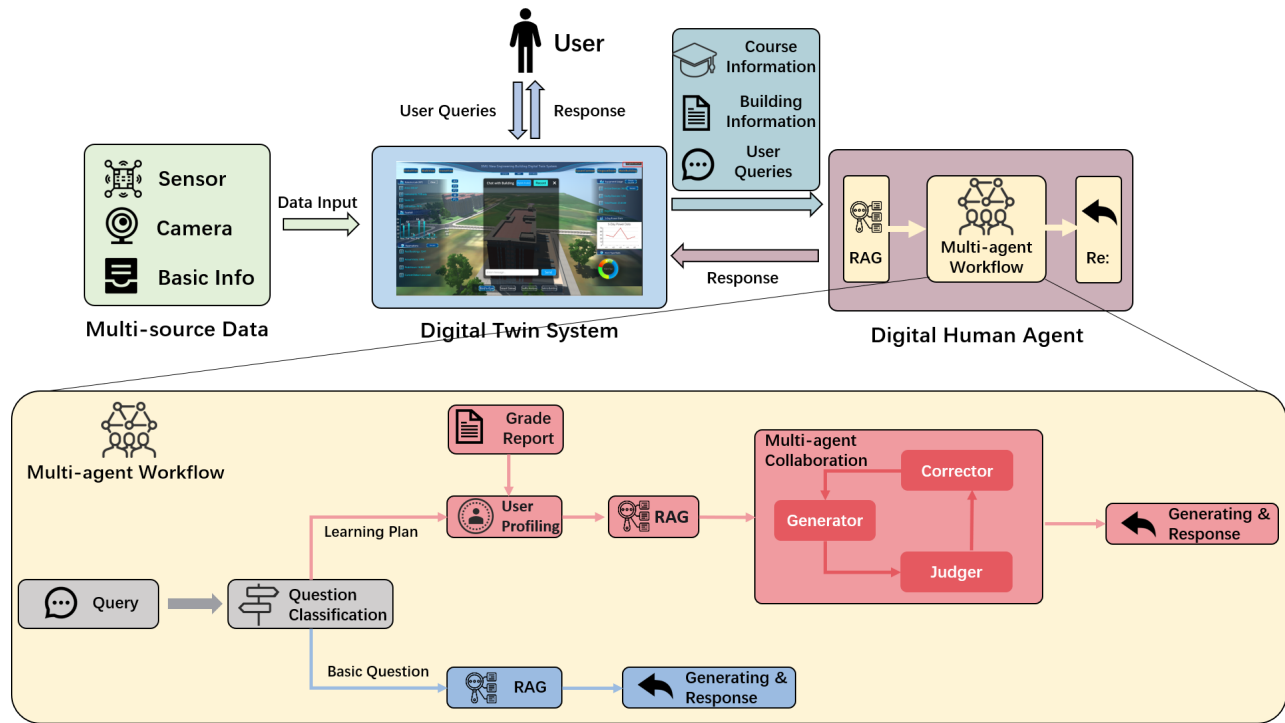


Figure 1: System Framework of Emerging Engineering R&D Building Digital Twin System

system integrates data from various sources, providing an information foundation for the intelligent agent to generate responses and interact with users.

2.2 Multi-agent Framework

For students and practice base managers, we have developed a digital human based on a multi-agent collaboration framework, in addition to the digital twin system. This digital human can grasp the basic information and operation of the practice base, as well as the courses and lab equipment related to Emerging Engineering Education. It can accurately answer users' questions about the practice base. In addition, it can serve as a student academic advisor in Emerging Engineering Education. Students can share their desired research directions with the digital human, which then analyzes their academic performance data to create personalized learning plans. These plans can interact with the digital twin system, enabling students to efficiently engage in learning and practice activities in the practice base according to their learning plans.

As single large model has limited context window[9], it is hard to handle diverse user queries. To solve this limitation, we have designed a multi-agent framework based on triage-style classification. It follows a user profiling, plan generation, reflection, and iteration approach, offering a new way to build digital human agents.

Triage. After the user asks a question, it is first analyzed by a triage-style classifier to determine the intent. Based on this analysis, the question is routed to the appropriate workflow path, improving the accuracy and efficiency of the response.

User Profiling. For complex questions about academic planning, a profiling agent first creates an academic profile of the student based on their learning needs and academic performance. This ensures that the subsequent plan-generation agent can quickly capture key information and produce high-quality academic plans.

Plan Generation. After extracting the relevant student information, the "generator" agent uses RAG to search for courses and lab equipment available in the practice base. The retrieved information, combined with the student profile, is fed into a large model to generate a learning plan. This plan includes courses arranged by difficulty and recommended lab equipment.

Reflection and Iteration. The initial learning plan is evaluated by an "judger" agent. It checks the plan against the student profile and teaching plan to ensure practicality and reasonableness. If problems are found, the "judger" agent suggests improvements and the "corrector" agent updates the plan. In addition, users can provide feedback on the learning plan, and the "generator" agent can revise it based on their input.

The learning plan can interact with the digital twin system. For equipment-searching users, they can click the link button in the generated plan, and the digital twin system will show its location in the practice base for easy booking.

The collaboration of the digital twin and the digital human systems fully utilizes the information generated by the practice base. It helps managers understand the operation of the base and make informed decisions. For students, it facilitates their use of the Emerging Engineering practice base, promoting innovation and practical activities under the Emerging Engineering education system.

3 Demonstration

In the demonstration part, we presented a digital twin system and a digital human agent built on the basis of the emerging engineering practice base. The system shows equipment usage, course participation, and allows infrastructure operation control. Users can question the digital human to understand the building's status. For students, proper academic planning and guidance are crucial. Students can state their learning needs to the digital human, then the agent creates a customized academic plan through profiling, generating, reflecting, and iterating. Multi-round dialogues allow the digital human to adjust plans according to the student's needs. Finally, students can use the digital twin system to quickly locate relevant classrooms and equipment via provided links, making it easier to carry out their learning and practice plans.

As shown in Fig.2, we compared the academic plans generated by single-agent and multi-agent systems. Fang Mingyang exhibits a pronounced interest in robotics and has achieved outstanding performance in foundational courses such as circuit design and programming. Both frameworks tailored their curricula to his profile; however, the single-agent schedule includes “Comprehensive Experiment in Environmental Science”, a course with limited relevance to robotics, whereas the multi-agent framework correctly produced a curriculum that is strictly aligned with robotic topics. The multi-agent framework produces more student-tuned and accurate curriculum plans. However, the single-agent framework provides insufficient guidance on course sequencing and the precautions required for laboratory equipment, which may lead students to operate instruments without adequate preparation, thereby introducing safety hazards in the lab. Overall, we find that the multi-agent

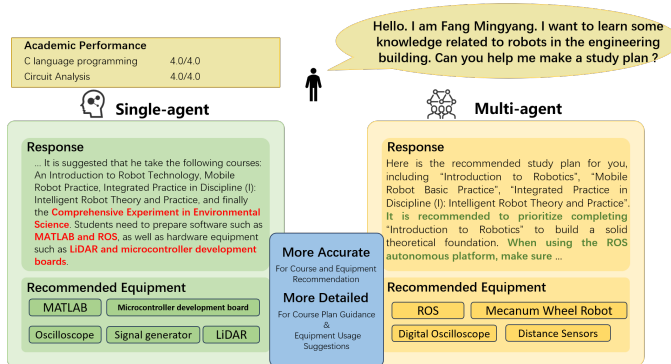


Figure 2: An example of Learning Plan Generation for Single-agent and Multi-agent

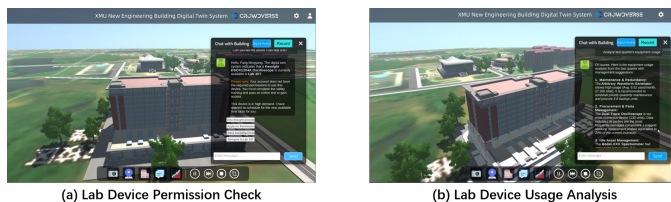


Figure 3: Lab Management Cases

framework yields learning plans that are both more aligned with individual learner needs and more accurately scoped than those produced by a single agent, while also offering novices markedly finer-grained guidance and recommendations.

In the laboratory management phase, as shown in Fig.3, students must pass a permission check before using any recommended equipment. The agent notifies those lacking authorization to apply for access prior to starting their plan. Since permission requires completing training and passing an exam, the process guarantees safe execution of the plan while equipping students with proper operational knowledge. Furthermore, the agent analyzes device-usage frequency to provide managers with targeted maintenance or replacement recommendations, thereby enhancing utilization efficiency and extending equipment lifespan.

4 Future Work

This demonstration shows how, in the Emerging Engineering education system, digital twins of practice bases and digital humans can interact. It improves the use of practice base information and offers new models for building similar education practice bases and apps.

In the future, we aim to fully use the data from practice bases to develop an intelligent manager. It can adjust management strategies automatically based on the base's operation for managers' reference or automated management. We will also create algorithms to assign tasks reasonably to smart robots in the building, helping managers automate property management.

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